# Chapter 2 : Examining relationships

03 June 2024 11:26



Prelude

**How much for my house?**

As the mortgage officer for the local bank branch, Matthew helps individuals make decisions about buying and selling houses. A couple planning to buy a new house wonders how much their current house will sell for. The city assessor's office provides *assessed values* for all houses in town for tax purposes. An assessed value is an estimate of the value of a home used to determine property taxes. Matthew knows that most realtors in town tell someone trying to sell a house that they should expect to sell the house for approximately 10% above its assessed value, but he isn't sure about this "rule of thumb."

What is the relationship between selling price and assessed value for houses in town? Matthew took an introductory business statistics course in college, and he recalls studying relationships between two variables like selling price and assessed value. A trip to the city assessor's office provides a sample of the selling prices and assessed values for recently sold houses. A scatterplot shows that the relationship between selling prices and assessed values can be described by a line with a positive slope of approximately 1.07. The value of the slope indicates that selling prices are about 7% above assessed values in Matthew's sample of recently sold houses. While some houses did sell for 10% above their assessed value, others sold for smaller percentages above assessed. Based on actual sales data, the slope of Matthew's line will provide his customers with a more reasonable assumption about selling prices compared to the "optimistic" 10% rule of thumb the realtors use.

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⇒ A marketing study finds that men spends more money online than women.

⇒ An insurance group reports that heavier cars have fewer deaths per 10,000 vehicles registered than do lighter cars.

⇒ These and other such statistical studies look at relationship between two variables.

- Response Variable, Explanatory Variable :-

⇒ A response variable measures an outcome of a study.

⇒ An explanatory variable explains or influences changes in a response variable.

⇒ Often times response variable is called the dependent variable, and explanatory variables is called independent variables.

- Example : The Just price?

⇒ The sellers profit for an item increases as the price is reduced, due to increased sales, until the point at which lower profit per item cancels out the rising sales.

⇒ A retail chain therefore introduces a new DVD player at several different price points and monitor sales.

↳ The chain wants to discover at which price its profits are greatest.

⇒ Hence total profit from sales of DVD player is the response variable, and the price is the explanatory variable.

- Scatterplots :

## 2.1 Scatterplots

CASE 2.1

### SALES AT A RETAIL SHOP

Three entrepreneurial women open Duck Worth Wearing, a shop selling high-quality secondhand children's clothing, toys, and furniture. Items are consigned to the shop by individuals who then receive a percentage of the selling price of their items. Table 2.1 displays daily data for April 2000 on sales at Duck Worth Wearing.[1] Larger retail establishments keep more elaborate records, particularly relating to sales of specific items or branded lines of goods. The consignment shop does not sell multiple copies of the same item, though records of sales by categories of item (clothing, toys, and so on) might be useful for planning and advertising.

The data in Table 2.1 concern each day's gross sales, broken down by method of payment (cash, check, credit card). "Gross sales" in retailing are overall dollar sales, not adjusted for returns or discounts. The data record the number of items sold and the dollar amount of gross sales for each payment method on each day. We will examine relationships among some of these variables.

The most common way to display the relation between two quantitative variables is a *scatterplot*. Figure 2.1 is an example of a scatterplot.

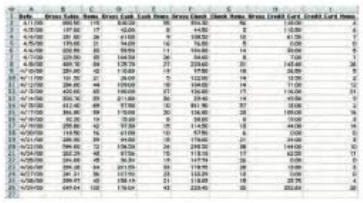


**TABLE 2.1** Gross sales and number of items sold.

⇒ The most common way to display a relationship between two quantitative variables is a scatterplot.

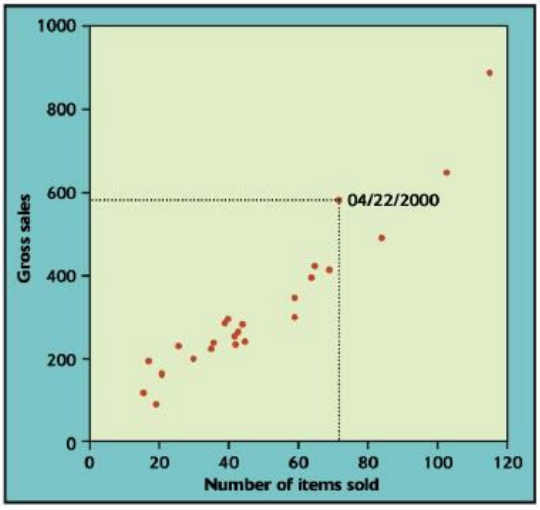



**FIGURE 2.1** Scatterplot of the gross sales for each day in April 2000 against the number of items sold for the same day, from Table 2.1. The dotted lines intersect at the point (72, 594), the data for April 22, 2000.

□ Items sold & gross sales :-

⇒ The figure (scatter plot) above shows scatterplot between items sold, gross sales.

↳ The number of items sold should explain gross sales, hence "items sold" is the explanatory variable, and "gross sales" is the response variable.

⇒ We want to see how gross sales change when number of item sold changes, so we put number of items sold on the horizontal axis.

⇒ A scatterplot shows the relationship between two quantitative variables measured on

the same individuals.

- Always plot the explanatory variable, if there is one, on the horizontal axis (the x-axis) of a scatterplot.
  - We usually call the explanatory variable x & the response variable y.
  - If there is no explanatory-response distinction, either variable can go on horizontal axis.

- Examining a scatterplot :-

⇒ In any graph, look for overall pattern, and for striking deviations from that pattern.

⇒ You can describe the overall pattern of the scatterplot by the form, direction, and strength of the relationship.

⇒ An important kind of deviation is an outlier, an individual value that falls outside the overall pattern of the relationship.

- Positive association, negative association :-

⇒ Two variables are positively associated when the above-average values of one tend to accompany above-average value of the other, and below average values also tend to occur together.

⇒ Two variables are negatively associated when above-average values of one tend to accompany below-average value of the other, and vice-versa.

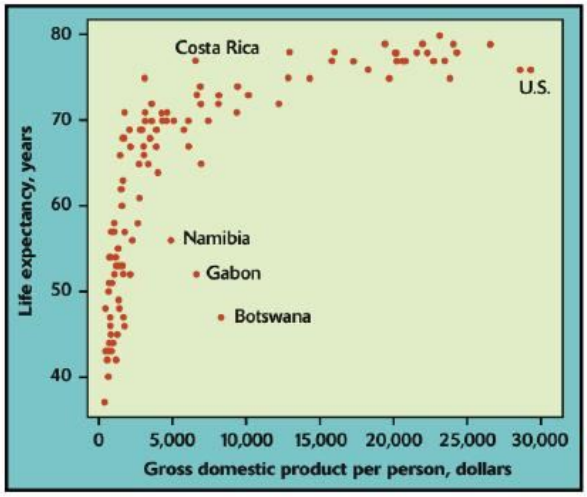



**FIGURE 2.7** Scatterplot of life expectancy against gross domestic product per person for all the nations for which data are available, for Exercise 2.10.

Not all relationship have a clear direction that we can describe as positive association or negative association. The figure 2.7 gives an example that is not linear and

has no clear direction.

- Adding categorical variables to scatterplots :-

⇒ Every business has time of the days, days of the weeks, and periods in the year when sales are higher.

□ ex:- are saturdays different?

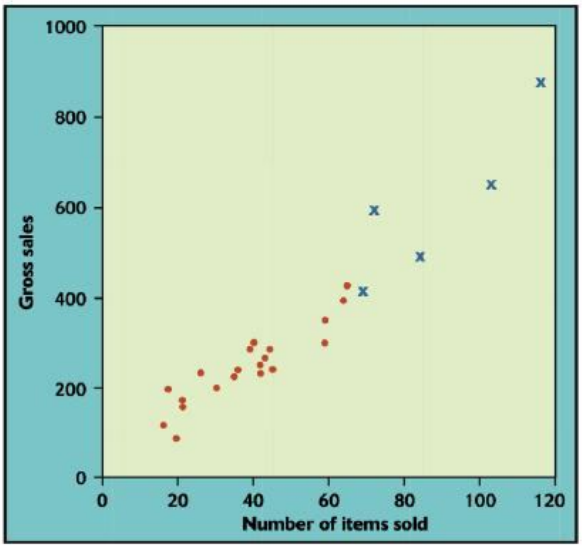



**FIGURE 2.4** Gross sales at Duck Worth Wearing against number of items sold, with Saturdays highlighted.

⇒ The above figure enhances the Figure. 2.1 by plotting the saturdays with a different plot symbol ( x ).

↳ The five days with the highest number of items sold are the five Saturdays in April 2000.

⇒ Dividing days into "Saturday" and "week day" introduces a third variable into the scatterplot.

↳ This is a categorical variable that has only two values.

⇒ Use different colors or symbol to plot points when you want to add a categorical variable to a scatterplot.

- Correlation :-

⇒ A scatterplot displays the form, directions, and strength of the relationship between two quantitative variables.

⇒ We say a linear relation is strong if the points lie close to a straight line, and weak if they're widely scattered about a line.

⇒ Our eyes are not good judges of how strong a linear relationship is.

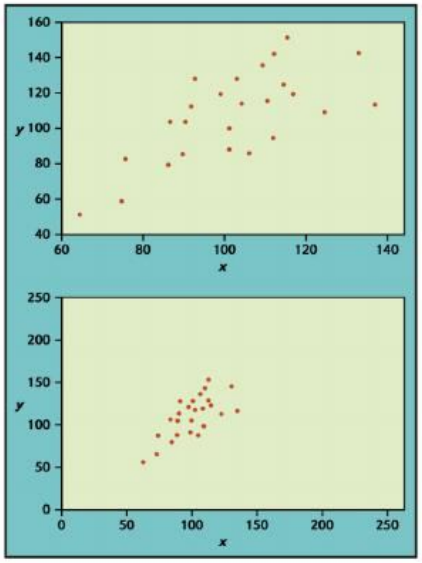



**FIGURE 2.9** Two scatterplots of the same data. The straight-line pattern in the lower plot appears stronger because of the surrounding open space.

⇒ The above two scatterplots depict the same data, but the lower plot is drawn smaller in a larger field.

↳ Yet the lower plot seem to have strong relationship.

↳ Hence, we need to follow our strategy for data analysis by using a numerical measure to supplement the graph.

↳ We use correlation as a numerical measure.

◦ The correlation r :-

⇒ The correlation measures the direction & strength of the linear relationship between two quantitative variables. It's usually written as r.

⇒ The correlation r on variables x and y for n individuals is :-

$$r = \frac{1}{n-1} \sum \left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

Where $\bar{x}$ & $\bar{y}$ are the means & $s_x$ & $s_y$ is the standard deviation.

⇒ The formula for r begins by standardizing the observations.

↳ The standardized value (z-score) says how many std above or below the mean the value lies.

◦ Facts about correlation :-

1) Correlation makes no distinction between explanatory & response variables.

2) Correlation requires both variables to be quantitative, so that it makes sense to do arithmetic indicated by the formula for r. (We cannot calculate correlation between income group of people & the city name they live in, because city is a categorical variable).

3) Because r uses standardized values, r doesn't change when we change the unit of measurement.

4) Positive r indicates positive association between variables, & negative r indicates negative association.

5) The correlation r is always a number between -1 and 1.
   - Values of r near 0 indicate a weak linear relationship.
   - r close to +1 or -1 indicates that the points in a scatter lies close to a straight line.

6) Correlation r measure the strength of linear relationship only between two variables, and doesn't show curved relationships, no matter how strong they are.

7) Like mean & standard deviation, the correlation is not resistant: r is strongly affected by a few outlier.

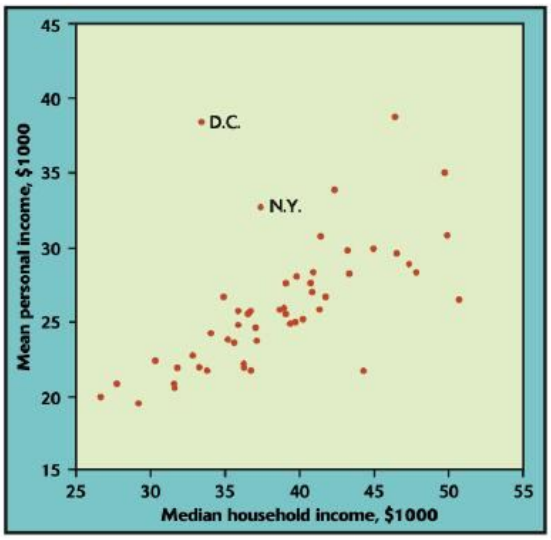



**FIGURE 2.8** Scatterplot of mean income per person versus median income per household for the states, for Exercise 2.11.

8) The correlation for the figure 2.8 is r = 0.6287 when all 51 observations are included but rises to r = 0.7789 when we omit Alaska & District of Columbia.
   - Use r with caution when outlier appears in scatterplot.



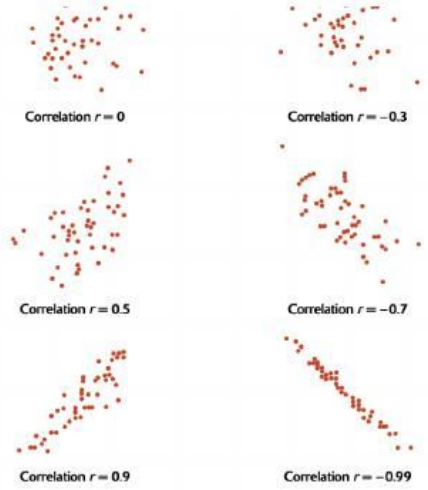



**FIGURE 2.10** How correlation measures the strength of a linear relationship. Patterns closer to a straight line have correlations closer to 1 or −1.

The standard deviations of both the variables are equal & horizontal & vertical scales are the same.

⇒ Correlation is not a complete description of two variable data, even when the relationship between the variable is linear. You should gives means & standard deviations for both x and y along with correlation.

↳ Because correlation uses mean & std, these measures are the proper choice to accompany the correlation.

◦ Least - Squares Regression :

⇒ If a scatter plot shows a linear relationship between two variables, we would like to summarize this overall pattern by drawing a line on scatter plot.

⊗ Regression line : A regression line is a straight line that describes how a response variable y changes as an explanatory variable x changes.

↳ We often use a regression line to predict the value of y for a given value of x.

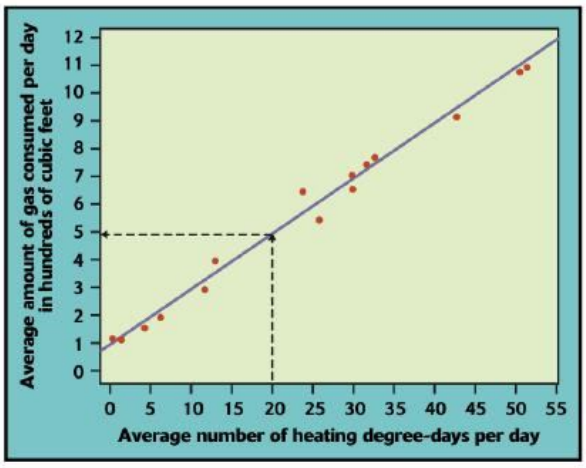



**FIGURE 2.11** The Sanchez household gas consumption data, with a regression line for predicting gas consumption from degree-days. The dashed lines illustrate how to use the regression line to predict gas consumption for a month averaging 20 degree-days per day.

⇒ In the above figure with regression, the Sanchez household can use it to answer question like: "If a month average 20 degree-days per day how much gas will we use?"

↳ To predict gas consumption, first locate 20 on the x-axis & then go up to the line to find the corresponding y on the line.

◦ The least-squares regression line :-

⇒ Well use line to predict y from x, so the prediction errors we make are errors in y, the vertical direction in scatterplot.

∴ error = observed-y - predicted-y.

⇒ No line will pass exactly through all point in scatterplot, but we want the vertical distances of the points from line to be as small as possible.

↳ There are many ways to make collections of vertical distances (errors) "as small as possible". The most common method is least-squares method.

◦ Least-squares regression line :-

⇒ The least-squares regression line of y on x is the line that make the sum of squares of vertical distances of the data points from the line as small as possible.

⇒ We can give the recipe for least squares in terms of means, and standard deviations of two variables & their correlations.

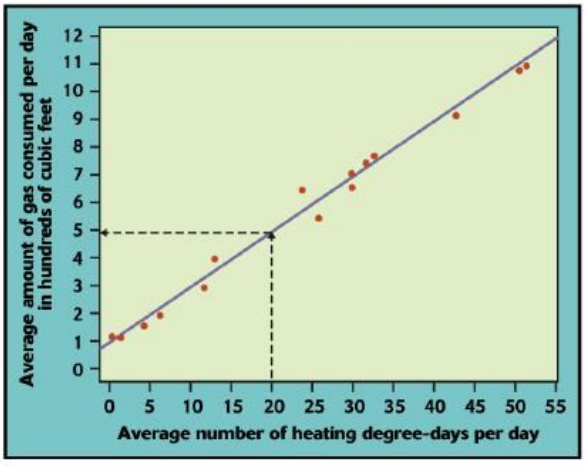



**FIGURE 2.11** The Sanchez household gas consumption data, with a regression line for predicting gas consumption from degree-days. The dashed lines illustrate how to use the regression line to predict gas consumption for a month averaging 20 degree-days per day.

◦ Equation of the least square regression line :-

⇒ We have data on an explanatory variable $x$ and a response variable $y$ for $n$ individuals.

↳ From the data calculate the means $\bar{x}, \bar{y}$ and the standard deviations $s_x$ and $s_y$, and their correlation

The least square regression line is then defined as :-

$$\hat{y} = a + bx$$

with slope $b = r \frac{s_y}{s_x}$.

& intercept

$$a = \bar{y} - b\bar{x}.$$

- Using a regression line :

⇒ The slope ($\overset{\text{slope}}{(a+b)}$) of a regression line is always important for interpreting the data.

↳ The slope is the rate of change, the amount of change in $y$ when the $x$ increases by 1.

intercept : The intercept of the regression line is the value of $y$ when $x = 0$.

↳ It's statistically meaningful only when $x$ is near 0.

- Facts about least-squares regression :

Fact 1 The distinction between explanatory and response variable is essential in regression.

If we reverse the role of the two variables, we get a different least-squares regression line.

↳ Hence you must decide which variable is response & which is explanatory.

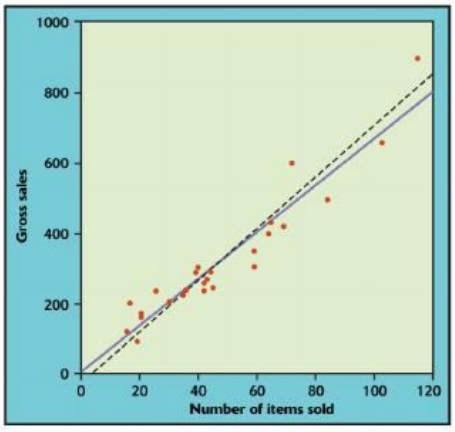



**FIGURE 2.14** Scatterplot of the gross sales at Duck Worth Wearing for each day in April 2000 against the number of items sold for the same day, from Table 2.1. The two lines are the two least-squares regression lines: of gross sales on item count (solid) and of item count on gross sales (dashed).

Fact 2. There is a close connection between correlation and the slope of the least-squares line. The slope is

Fact - 2: There is a close connection between correlation & the slope of the least-squares line.

$\therefore y = a + bx$ (slope)

& $\therefore$ slope $b = r\frac{s_y}{s_x}$

The equation of b states that a change of one standard deviation in x corresponds to a change of r standard deviations in y along the regression line.

→ When the variables are perfectly correlated ($r = 1$ or $r = -1$) the change in y is same as change in x.

Fact 3: The regression line always passes through the point $(\bar{x}, \bar{y})$ on graph of y against x.

→ Here $\bar{x}, \bar{y}$ is the mean of x & y respectively.

→ So the regression line is the line with slope $r\frac{s_y}{s_x}$, which passes through $\bar{x}$ & $\bar{y}$. Hence we can represent the regression line in terms of $\bar{x}, s_x, \bar{y}, s_y$, and r.

Fact 4: The correlation r describes the strength of a straight line relationship.

→ In regression setting this description takes a specific form: the square correlation $r^2$ is the fraction of the variation in values of y that is explained by the least-squares regression of y on x.

$\therefore r^2 =$ variation in y as x pulls it along the line / total variations in observed values of y.

□ Using $r^2$ :-

⇒ The correlation between degree days & gas use is $r = 0.9953$. So $r^2 = 0.99$ & the straight line relationship explains 99 % month to month variations in gas use.

⇒ In figure 2.14, $r = 0.9540$ & $r^2 = 0.91$. The linear relationship between gross sales and item count explains 91% of the variation in either variable.

→ There are two regression lines but just one correlation, and $r^2$ helps interpret both regressions.

When you report regression, give $r^2$ as a measure of how successful the regression was in explaining the response.

→ All software outputs include $r^2$ either as decimal, or as percentage.

○ Residuals :-

⇒ Deviations from overall result is also important.

⇒ A residual is the difference between an observed value of response variable and the value predicted by the regression line.

i.e residual = observed $y$ - predicted $y$

= $y - \hat{y}$ ($\hat{y}$ here denotes prediction).

□ Examples: Predicting credit card sales :-

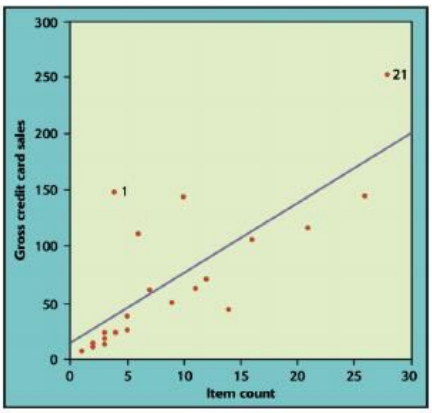



**FIGURE 2.15** Scatterplot of gross credit card sales at Duck Worth Wearing versus number of items purchased with a credit card for 21 days, from Table 2.1. The line is the least-squares regression line for predicting sales from item count.

⇒ The above plot shows a positive association - days with more items purchased with a credit card tend to be the days with higher credit card sales.

↳ The overall pattern is moderately linear. ($r = 0.7718$)

⇒ There is a residual for each data point.

⇒ The residual from least square line have a special property: the mean of the least squares residuals is always zero.

Residual plot: It's a scatterplot of the regression residuals against the explanatory variable.

↳ Residual plots helps us assess the fit of a regression line.

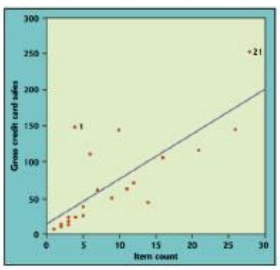



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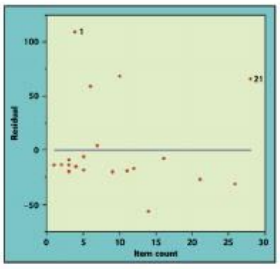



**FIGURE 2.16** Residual plot for the regression of sales on item count. Observation 1 is an outlier.

Observation 21 has a smaller residual but is more influential.

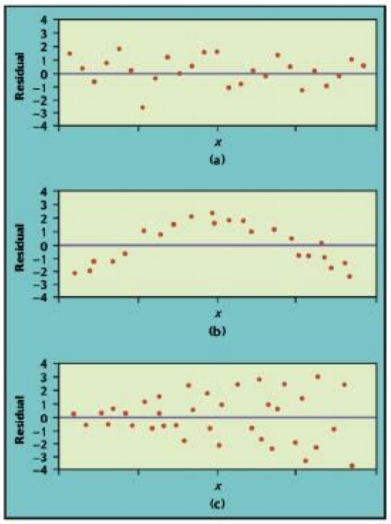



**FIGURE 2.17** Idealized patterns in plots of least-squares residuals. Plot (a) indicates that the regression line fits the data well. The data in plot (b) have a curved pattern, so a straight line fits poorly. The response variable $y$ in plot (c) has more spread for larger values of the explanatory variable $x$, so prediction will be less accurate when $x$ is large.

⇒ If the regression line captures the overall relationship between x & y, the residuals should have no systematic pattern ex:- 2.17(a).

⇒ Below are few points to consider if you examine the residuals using either a scatterplot of the data, or a residual plot.

→ A curved pattern (Figure 2.17(b)) shows relationship is not linear, hence a straight line is not a good summary for such data.

→ Increasing or decreasing spread about the line as x increases (Figure 2.17(c)).

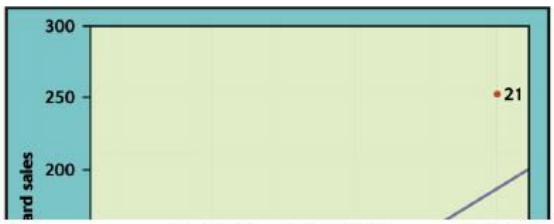
↳ Predictions of y will be less accurate for larger x in that example.

→ Individual points with larger residuals: such points are outliers in the vertical (y) direction, because they lie far from the line that describes overall pattern.

→ Individual points that are extreme in the x direction (like observation 21 in Fig 2.15 & 2.16)

↳ Such points may not have large residuals, but they can be very important. We address such points next.

- Influential observations :-



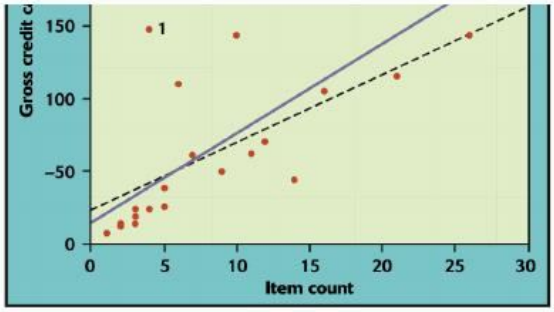



**FIGURE 2.18** Two least-squares regression lines of sales on item count. The solid line is calculated from all the data. The dashed line was calculated leaving out observation 21. Observation 21 is an influential observation because leaving out this point moves the regression line quite a bit.

⇒ Observations 1 and 21 are both unusual in the above figure

→ Observation 1 lies far from the regression line.

→ Observation 21 is close to the line, but far out in the x direction.

→ Because of its extreme position on the item count scale (x-axis), this point has a strong influence on the position of the regression line, we call such points Influential.

→ The above figure adds a second regression line, calculated after leaving observation 21. The line moves quite a bit.

⇒ Influential observations may have small residuals, because they pull the regression line towards themselves.

A observation is called influential for a statistical calculation if removing it would markedly change the result of the calculation.

→ Points that are extreme in the x direction of a scatterplot are often influential for the least square regression line.

An influential observation :-

⇒ The strong influence of 21 makes the original regression misleading.

→ The original data have $r^2 = 0.5957$. (about 60% of variation is explained by least squares).

→ If we leave out 21, $r^2$ drops to 43%, so only 43% of data is explainable.

Q) What should we do with 21?

→ If this unusual day resulted from special promotion, we might remove it when predicting for ordinary business days.

Cautions about correlation and Regression :-

Correlation and regression only describe linear relationship.

The correlation and least-squares regression line are not

resistant.

- Beware extrapolation:

⇒ Association for variables can be trusted only for the range of values for which the data is collected.

↳ Even a very strong relationship may not hold outside the data's range.

**EXAMPLE 2.15 Predicting the future**

Here are data on the number of Target stores in operation at the end of each year in the early 1990s:[21]

| Year ($x$) | 1990 | 1991 | 1992 | 1993 |
|---|---|---|---|---|
| Stores ($y$) | 420 | 463 | 506 | 554 |

These points lie almost exactly on a straight line, with $r = 0.9996$. The least-squares regression line for these data predicts 864 Target stores in operation by the end of 2000. However, the Target 2000 Annual Report shows 977 stores in operation at the end of 2000. The very strong linear trend evident in the 1990–1993 data did not continue to the year 2000.

⇒ Extrapolation is the use of a regression line for a prediction far outside the range of values of the explanatory variable $x$ that you used to obtain the line.

↳ Such predictions are often not accurate.

- Beware correlations on averaged data:-

⇒ Correlations based on averages are usually too high when applied to individuals

↳ ex:- Averaging over a month of gas consumed smooths out day to day variations like doors left open, house guests using more gas & so on.

- Beware the lurking variable:-

⇒ Often relation between two variables is strongly influenced by other variables.

⇒ A lurking variable is a variable that is not among the explanatory or response variables in the study and yet they may influence the interpretation of relationships among those variables.

↳ It can falsely suggest a false strong relationship, or it can hide a relationship that is really there.

□ Example: Discrimination in medical studies:

⇒ Studies shows that men who complaint of chest pain are more likely to get detailed tests and aggressive treatments such as bypass surgery than are woman with similar complaints

↳ Is this association between gender & treatment due to discrimination.

⇒ Perhaps not. Men & Women develop heart problems at different ages: women are on the average between 10 and 15 years older than men.

→ Aggressive treatment are more risky for older patients, so doctors may hesitate to recommend them.

→ Here the lurking variables are the patient's age, and condition may explain the relationship between gender & doctors decision.

◦ Association is not causation:

⇒ A strong association is not enough to draw conclusions about cause & effect.

□ Example: Does television extend life?

⇒ There is a high positive correlation: nations with many TV sets have higher life expectancies.

⇒ The basic meaning of causation is that by changing x we can bring a change in y.

→ Could we lengthen the lives of people in Rwanda by shipping them TV sets?

→ No, rich nations have more TV sets, & also have a better life expectancies, because they offer better nutrition.

→ There is no cause & effect tie between TV sets & length of life.

Q) What are the criteria for establishing causation, when we cannot do an experiment?

1) The association is strong: The association between lung cancer & smoking is very strong.

2) The association is consistent: Many studies of different kinds of people in many countries link smoking to lung cancer.

→ That reduces the chances of a lurking variables specific to one group or one study.

3) Higher Doses are associated with stronger response
- People who smoke over a longer period get lung cancer more often, and people who stop smoking reduce their risks.

4) The alleged cause precedes the effect in time
- Lung cancer develops after years of smoking. The number of man dying of lung cancer rose as smoking became more common, with a lag of about 30 years.

5) The alleged cause is plausible
- Experiments on animals have shown that tars from cigarette smoke do cause cancer.

- Relations in Categorical Data :

⇒ To analyze categorical data we use counts or percents of individuals that fall into various categories.

Case 2.2 : Marital Status & Job Level :

⇒ We sometimes read in magazines that getting married is good for your career.

⇒ The below table have grouped the many job grades into quarters.

↳ Grade 1 contains jobs in the lowest quarters of the job grades, and grade 4 contains those in higher quarter.

**TABLE 2.10 Marital status and job level**

| | Marital status | | | | |
|---|---|---|---|---|---|
| Job grade | Single | Married | Divorced | Widowed | Total |
| 1 | 58 | 874 | 15 | 8 | 955 |
| 2 | 222 | 3927 | 70 | 20 | 4239 |
| 3 | 50 | 2396 | 34 | 10 | 2490 |
| 4 | 7 | 533 | 7 | 4 | 551 |
| Total | 337 | 7730 | 126 | 42 | 8235 |

- Marginal distributions :-

Q) How can we best group the information contained in table 2.10?

⇒ First look at the distribution of each variable separately.

↳ The distributions of job grade alone or marital status alone are called marginal distributions.

- Calculating a marginal distribution :-

⇒ The percentages of man who have grade 1 job is

$$\frac{\text{total with grade 1}}{\text{table total}} = \frac{955}{8235} = 0.1160 = 11.6\%$$

Like that we obtain marginal distribution of job grade in percent

| Job grade | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| Percent | 11.6% | 51.4% | 30.2% | 6.7% |

⇒ The total percent should be 100%. Here it is 100.01% due to round off error.

⇒ We can use bar or pie-chart to display such distribution.

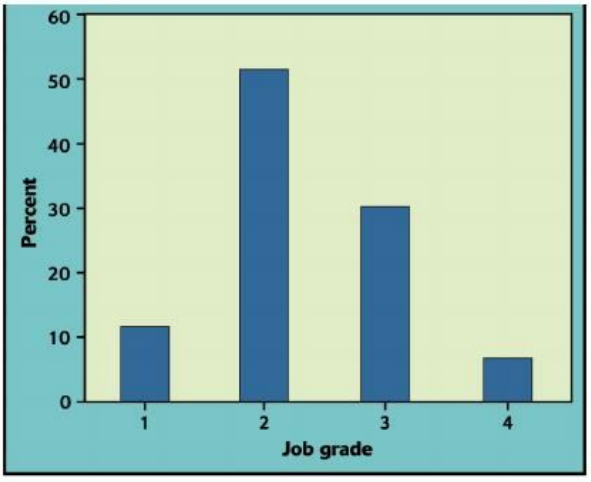



**FIGURE 2.22** A bar graph of the distribution of job grade for the men described in Case 2.2. This is one of the marginal distributions for Table 2.10.

- Describing Relationships :-

⇒ Table 2.10 contains much more information than the two marginal distributions alone.

⇒ Relationships among categorical variables are described by calculating appropriate percents from the counts given.

- Example 2.2: How common are grade 1 jobs?

Q) What percentage of single men have grade 1 jobs?

⇒ This the count of those who are single and have grade 1 jobs as percent of the single group total:

$$\frac{58}{337} = 0.1721 = 17.21\%$$

⇒ In the same ways find the percent of men in each marital status group who have grade 1 jobs. The comparisions of all four groups is

| Marital status | Single | Married | Divorced | Widowed |
|---|---|---|---|---|
| Percent with grade 1 jobs | 17.21 | 11.31 | 11.90 | 19.05. |

- Conditional distributions :-

⇒ Example 2.2 doesn't compare the complete distributions of job grade in the four marital status groups.

↳ It compares only the percent who have grade 1 jobs. Let's look at the complete picture.

- Calculating a conditional distribution (ex: 2.23).

⇒

Marital Status

| Grade | Count / Col% | Single | Married | Divorced | Widowed | |
|---|---|---|---|---|---|---|
| | 1 | 58<br>17.21 | 874<br>11.31 | 15<br>11.90 | 8<br>19.05 | 955 |
| | 2 | 222<br>65.88 | 3927<br>50.80 | 70<br>55.56 | 20<br>47.62 | 4239 |

| Job G | | | | | | |
|---|---|---|---|---|---|---|
| | 3 | 50<br>14.84 | 2396<br>31.00 | 34<br>26.98 | 10<br>23.81 | 2490 |
| | 4 | 7<br>2.08 | 533<br>6.90 | 7<br>5.56 | 4<br>9.52 | 551 |
| | | 337 | 7730 | 126 | 42 | 8235 |

**FIGURE 2.23** JMP output of the two-way table of job grade by marital status with the four conditional distributions of job grade, one for each marital status group. The percents in each column add to 100% (up to roundoff error).

⇒ We use the term "conditional" because the column refers to only those men who satisfy a particular condition.

No single graph (such as scatterplot) portrays the form of the relationship between categorical variables.

↳ No single measure (such as correlation) summarizes the strength of the association.

⇒ For numerical reasons, we chose percentages.

↳ In this case you must decide which percentage you need.

↳ Compare the distributions of response variable (Job. grade) for the separate value of explanatory variable (marital status).

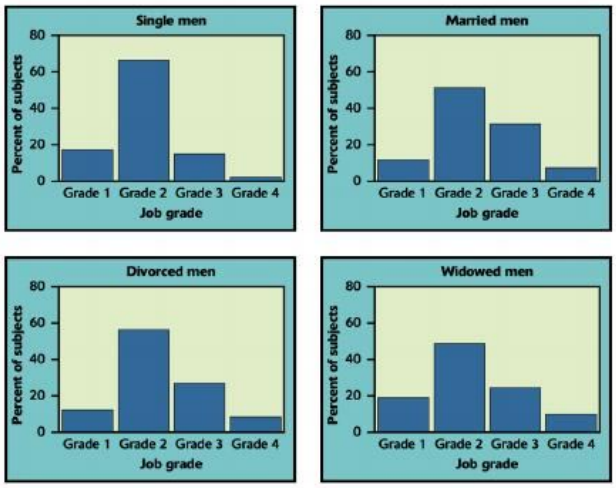



**FIGURE 2.24** Bar graphs comparing the distribution of job grade level within each of four marital status groups. These are conditional distributions of job grade given a specific marital status.